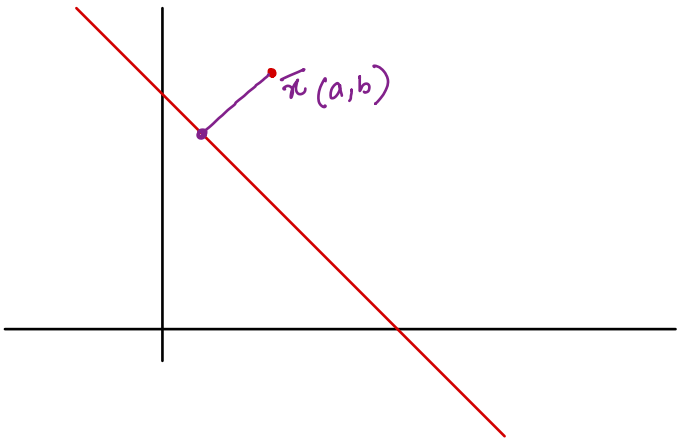


$$w_1x_1 + w_2x_2 + w_0 = 0$$

(a,b)

$$w_1(x_1 - a) + w_2(x_2 - b) + w_0 = 0$$

$$w_1x_1 + w_2x_2 + \underbrace{w_0 - w_1a - w_2b}_{w_0'} = 0$$

$$\bar{w}^T\bar{x} + w_0$$

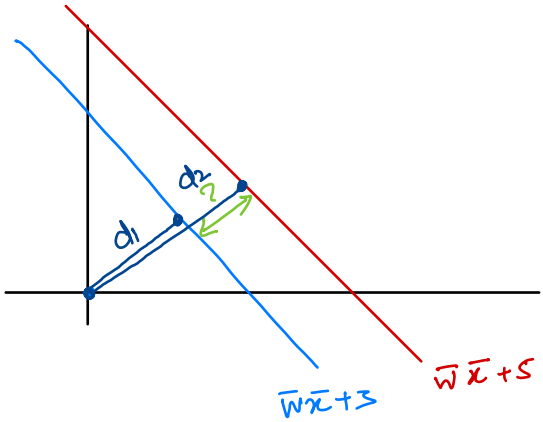
point i want to find dist of from the line

$$d = \frac{w_0'}{\|\bar{w}\|}$$

$$\bar{x} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$d = \frac{\bar{w}^T\bar{x} + w_0}{\|\bar{w}\|}$$

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

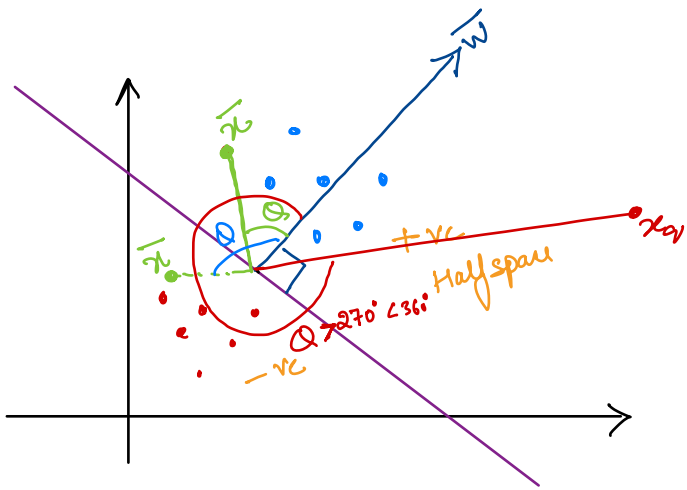


$$d_1 = \frac{w_0}{\|\bar{w}\|}$$

$$d_2 = \frac{w_0}{\|w\|}$$

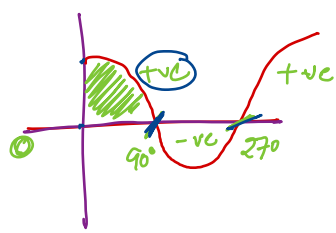
$$d_1 = \frac{3}{\|\bar{w}\|} \qquad d_2 = \frac{5}{\|\bar{w}\|}$$

$$d_2 - d_1 = \frac{5}{\|\bar{w}\|} - \frac{3}{\|\bar{w}\|} = \frac{2}{\|w\|}$$



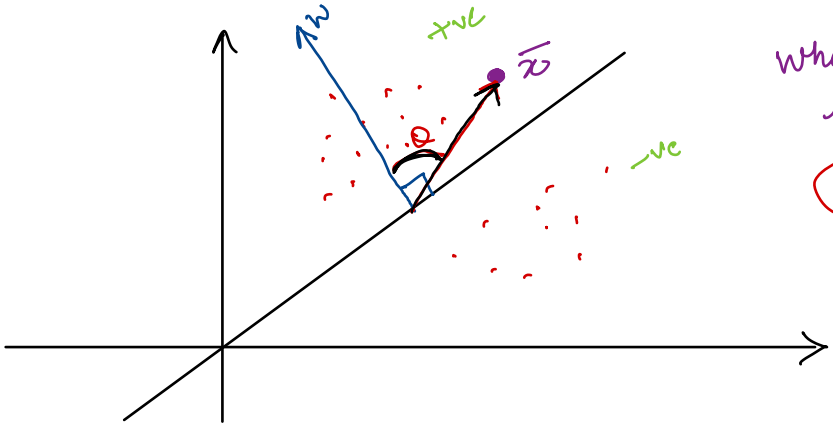
$\bar{w}\cdot\bar{x}$ = sign?

$= \|w\|\|x\|\cos\theta$



+ve position → θ b/w $\bar{x}$ & $\bar{w}$ is $0-90°$, $270-360°$

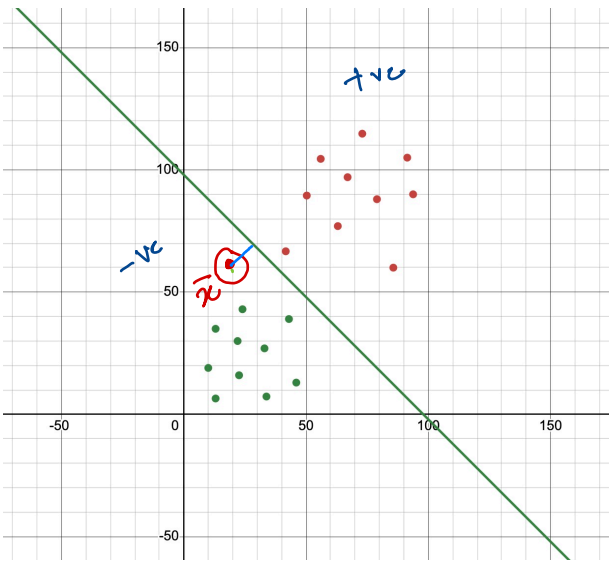
-ve → θ b/w $\bar{x}$ & $\bar{w}$ is $90-270$



Where is $\bar{x}$ lying?

+ve

Putting all of this together



$\bar{x}$ is mis-classified

Actual label = +1

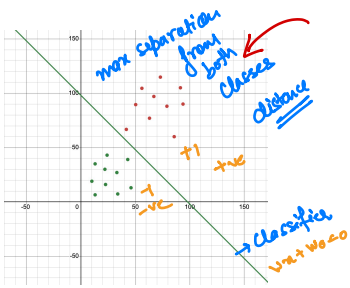
Assigned label = -1

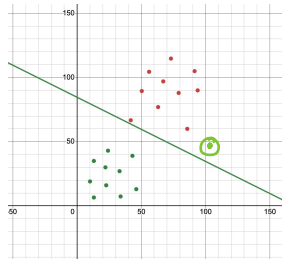
distance of a point from line

$$\left[d = \frac{w^T \bar{x} + w_0}{\|\bar{w}\|} \right]$$

$d < 0$

$d > 0$





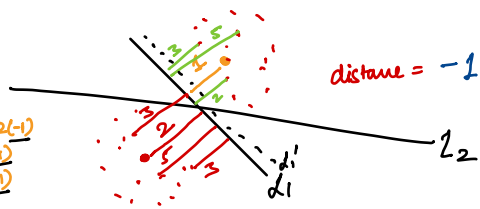
which line is better?

Maximise the distance b/w the points & the line

$$d = \sum_{i=1}^{n} \left(\frac{\bar{w}_1 \bar{x}_i + w_0}{\|\bar{w}\|} \right) y_i$$

$y_i = -1$ when -ve half space

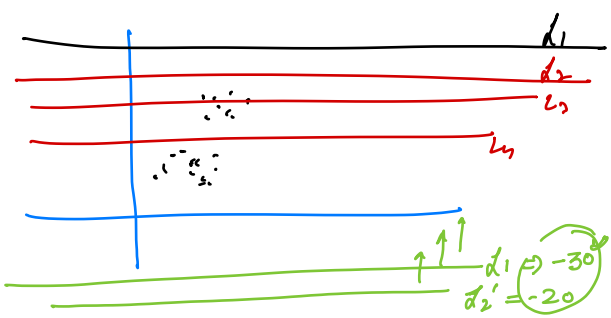
$y_i = +1$ +ve halfspace



$+1 - 2 \Rightarrow -1$

$d_1 = -20$ ✓ $L_2 = +10$ ✓

which is the better classifier?



$d_1 = 30$
$d_2 = 20$

Max Distance ↑

$-G \Rightarrow d_1 = 30 \rightarrow -30$
$d_2 \rightarrow 20 \rightarrow -20$

$-G \Rightarrow 30$
20

Max Distance → d_1 or d_2

⇓

Gain $= \sum \left(\frac{\bar{w}\bar{x}_i + w_0}{\|\bar{w}\|} \right) y_i$

$y_i = +1$

$y_i = -1$ (-ve half space)

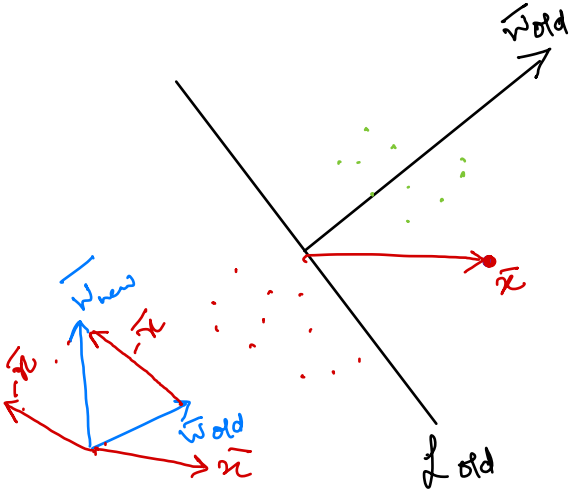
G Max Gain

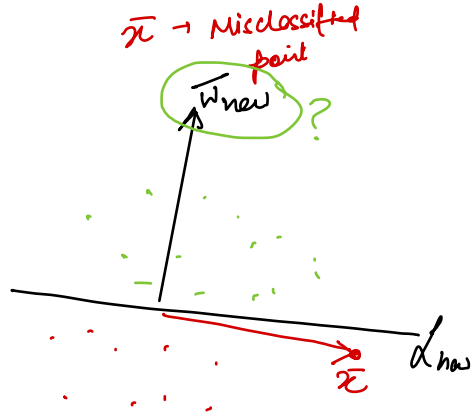
Loss $= -G$ Loss fnctn

Core terminology of Machine Learning

[Gradient Descent → what if your line is far, how would you correct.]

To end it all





In which direction should I move my line (clockwise or anticlock)

L_{old} $\quad w_1 x_1 + w_2 x_2 + w_0 = 0$

$2x_1 + 3x_2 + 5 = 0$

$L_{new} \rightarrow$ step change (w) would have changed

$$w^{new} = w^{old} + \underbrace{-\bar{x}}$$

Eq of new line ?

vector

In future [Gradient
↳ partial derivative

Gradient descent (Engine)

Bonus [if I do all of this in code ?]